

SAURAB MISHRA

Machine Learning | AI | Data Science | NLP | Technical Content Creator

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PROFESSIONAL SUMMARY

BS-MS student in Computational and Applied Mathematics at IISER TVM, working across AI research and engineering, spanning areas such as machine learning, large language models, knowledge-enhanced systems, agent-based architectures, and scalable model deployment. I'm contributing to research publications in NLP/IR and developing projects that bridge theory with practical impact. I share my learnings through technical writing and educational content, and I'm always open to collaborations.

EDUCATION

BS-MS Dual Degree, Computational and Applied Mathematics

Indian Institute of Science Education and Research (IISER), Thiruvananthapuram

2023 – 2028

India

RESEARCH EXPERIENCE

Research Intern Under Prof. Kripabandhu Ghosh

Indian Institute of Science Education and Research Kolkata

2024 – Present (1+ Year)

Remote

- Conducting advanced research in **Multi-Agent Systems (MAS)** for **Large Language Models (LLMs)**, focusing on novel architectural designs and inter-agent communication protocols
- Developing novel **fine-tuning methodologies** and **Natural Language Processing (NLP)** techniques to enhance model performance and mitigate biases in complex linguistic tasks
- Prepared and submitted **multiple research publications** currently under review at top-tier AI/ML conferences, demonstrating significant contributions to the field
- Contributing to cutting-edge research in **ethical AI** and **human-AI interaction**, exploring responsible AI development and user-centric design principles

MANUSCRIPTS UNDER REVIEW & IN PREPARATION

Manuscripts available upon request, subject to applicable review and anonymity policies.

1. **Exposure as a Behavioral Ranking Proxy: Validation under Fixed-Context RAG Replay**
Under double-blind review. Examines whether retrieval-stage exposure reliably ranks downstream attack success in RAG systems.
2. **Symbolic Prompt Schemas Change What Language Models Emit, Not What They Compute**
Under double-blind review. Studies the relationship between structured prompting, output format, and reasoning performance.
3. **Routing or Rationing? Decomposing Communication Control in Multi-Agent Language Models**
Under review. Investigates communication efficiency and the distinct roles of routing and message budgets in multi-agent LLM systems.
4. **When Does Multi-Agent Collaboration Help? Structured Decomposition for LLM-Based Data Annotation**
Under review. Examines when structured collaboration improves annotation quality across legal and bias-detection tasks.
5. **Ghost in the Machine**
In preparation. Explores parallel deliberative decoding and disagreement-based reliability signals in language models.
6. **ArguGenCourt: Courtroom Argument Generation via Judicial Multi-Agent LLMs**
Under review. Studies multi-agent approaches to legal argument generation and evaluation.

TECHNICAL CONTENT

Technical Blog Writing:

- **Medium Technical Blogger** - @saurabmishra under **Towards AI** publication
- Authored comprehensive technical articles on AI/ML topics, reaching readers worldwide
- Specialized content covering LLMs, RAG systems, and AI architectures

Educational Content Creation:

- Created in-depth **YouTube playlist** explaining **Retrieval-Augmented Generation (RAG)**
- Comprehensive technical tutorials bridging complex AI research with practical implementation

TECHNICAL SKILLS

Programming Languages and Relevant skills:

- Python, R, SQL, Matlab, C
- Data Science Techniques
- Advanced Algorithm Design

AI & Machine Learning:

- LLMs, Multimodal, RAG, RLHF
- Multi-Agent Systems
- Transformer Architectures, ML-Algorithms
- Fine-tuning & NLP

Frameworks & Libraries:

- TensorFlow, PyTorch
- Scikit-learn
- Hugging Face, LangChain

Cloud:

- AWS, Google Cloud, IBM Cloud, Azure

INNOVATIVE PROJECTS

Multi-Agent Reinforcement Learning for Emergent Coordination

Research Project (Ongoing)

- Investigating communication-efficient MARL algorithms for cooperative decision making.
- Developing decentralized agent coordination mechanisms using reinforcement learning.
- Evaluating scalability, emergent behaviors, and communication protocols in multi-agent environments.

Step Mentor: AI Educational Platform



Developed an adaptive learning platform using fine-tuned multimodal LLM, improving user retention through personalized study plans and integrated many features.

Research Project: Multi-Agent Systems for Enhanced LLM Performance



Enhance multi-agent architectures inspired by human cognitive processes to improve language model performance through adaptive collaboration and context-driven optimization. **Key Innovations:** Agent coordination, knowledge fusion, and dynamic contextual systems.

Research Project: AURA: Alignment Using Reinforcement and Human Assessment



AI training pipelines by incorporating human feedback to align systems with ethical standards and values. **Key Innovations:** Reinforcement learning algorithms and human-in-the-loop systems.

pRAGna: IISER Thiruvananthapuram Multimodal RAG System



Developed an open-source Retrieval-Augmented Generation (RAG) framework integrating text, PDF, and image understanding for institutional question answering and knowledge access.

- Designed a **hybrid retrieval pipeline** combining local semantic search (SentenceTransformers + ChromaDB) with a web-based fallback to rank and fetch information outside the knowledge base.
- Implemented **website chaining and ranking algorithms** for IISER TVM sources to ensure grounded, source-verified responses.
- Built a scalable backend using **Flask, LangChain, and LLMs** with a modern **React + TailwindCSS** frontend and smart caching mechanisms.
- Conducted prototype-level evaluation showing **query response times of 2–5s (simple) and 5–10s (complex)**, with web fallback adding +3–5s.
- Observed retrieval relevance of **90–95% (local), 85–90% (web fallback)**, leading to **95% combined accuracy** in internal testing.

- Demonstrated scalability up to **5,000 documents** (< 500ms retrieval latency) on local benchmarks, with future support planned for Pinecone/Weaviate vector DBs.

Multi Agent Job Search Automation Framework



A Multi-Agent System (MAS) framework for intelligent job search automation and application management, leveraging advanced AI techniques.

PROFESSIONAL CERTIFICATIONS

- **IBM Data Science Professional Certification** (2024) – Advanced Machine Learning Specialization
- **Generative AI with Large Language Models Certification** (2024) – LLMs Specialization
- **Neural Networks and Deep Learning** (2025) – Deep Learning Specialization
- **MCP Course: Advanced MCP Development** (2025)

CLUB AND EXTRA ACTIVITIES

Advisor (Current)

Coding Club of IISER TVM (CCIT)

Founder & Former Head; involved since 2024

- Advise the club on technical initiatives and mentor student project teams.
- Founded and grew the club to 200 active members, mentoring a team of 50 across three software projects.
- Developed the club website and delivered technical courses and hands-on project training.

RESEARCH INTERESTS

• Machine Learning • Large Language Models • Multi-Agent Systems • Retrieval-Augmented Generation • Reinforcement Learning with Human Feedback • Multi-Modals • Fine-tuning Methodologies • Natural Language Processing • Computer Vision • Human-AI Interaction